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Integrated Reservoir Studies Optimize Hydraulic Fracturing and Post-Frac Well Performance: Case Studies in Saudi Arabian Gas Sandstones

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Proposal

This paper presents a methodology for an integrated reservoir study to identify different facies in the reservoir, select perforations interval and type of hydraulic fracture, understand pre- and post-fracture reservoir performance, and optimize fracturing treatments and reservoir development. The target formation is mainly from a Permian eolian 'Unayzah-A gas bearing sandstone reservoir which is highly heterogeneous and poorly consolidated that requires a systematic and integrated assessment prior to its development.

Integrated reservoir studies utilizing engineering and petrophysical data showed the complexity and heterogeneity of the 'Unayzah-A reservoir. The reservoir constitutes of clay mineral (illite) and hence it is highly susceptible to formation damage associated with drilling fluids. In order to remove damage, control sand production, and improve productivity, all gas wells are routinely fracture stimulated employing screen or screenless completion.

This paper provides a summary of the reservoir and fracture treatments focusing particularly on the fracture design, execution, fracture modeling, and post-treatment production analysis from the recently completed wells. A comprehensive data set of logs, pre- and post-fracture pressure buildup tests (PBU's), drill stem tests (barefoot and cased hole DST), and detailed fracture modeling are presented to demonstrate the fracture performance in this complex eolian dune reservoir environment.

Estimation and interpretation of reservoir pressures, fracture geometry, and transient tests from these wells indicate possible water-zones compartmentalization, reservoir continuity, and depletion as primary drive mechanisms.

This paper shows that understanding reservoir quality and characteristics, using focused reservoir studies, leads to successful completion results and optimum development. The study identifies the dominant flow unit, eliminates unnecessary perforation intervals, determines stimulation treatment type and size and optimizes cost for overall field development.

Introduction

Approximately 70% of the world's oil and gas reserves are found in poor/weakly consolidated reservoirs.¹ The field of study, located in the southeastern part of the Saudi Arabia, comprises a deep 'Unayzah gas sandstone reservoir. Four production wells namely the W-1, W-5, W-6, and W-7 produce gas condensate from the Permian 'Unayzah A reservoir (Figure 1). Vertical and lateral variations of reservoir properties and unconsolidated sand zones have made the deep 'Unayzah reservoir development quite challenging to the reservoir management team. The synergistic effort at Saudi Aramco has managed to quickly address and overcome most of the challenges.¹⁻³

This paper discusses a deep gas eolian sandstone where good knowledge of reservoir quality and characteristics using focused reservoir studies led to successful completion results and optimum field development. Strategic reservoir assessment determines the dominant flow unit, eliminates unnecessary perforation intervals, determines treatment type and size, and optimizes cost for individual wells in particular and entire field in general.^{3,4}

The fracture treatment in Saudi Aramco includes fracpack (using mechanical screens) and screenless fracturing. Saudi Aramco normally conducts diagnostic tests and performance evaluations on the wells, prior to, during, and after fracture treatment to monitor the effectiveness of fracturing.

A brief summary of the pre- and post-fracture treatment from several gas wells is presented in this paper. A comprehensive data set of logs, pre- and post-fracture pressure buildup tests (PBU's), drill stem tests (barefoot and cased hole DST), and detailed fracture modeling were used to understand fracture performance in this complex eolian dune reservoir environment.

'Unayzah Reservoir Description

The 'Unayzah Formation is Permian (to Carboniferous) in age, and can be subdivided into three stratigraphic units. The lower (B and C) units are dominated by generally tight quartzose sandstones with rare siltstone intervals that are ascribed to deposition in a fluvial-dominated environment. Reservoir quality is commonly enhanced by natural fractures. The B unit in places contains deposits that have affinities with glacial diamictites. These two units are separated from the stratigraphically higher A unit by a disconformable surface. The 'Unayzah-A reservoir, defined by Area Exploration as 'Unayzah-A2, is composed of well-developed eolian sandstones and closely associated inter-dune and lacustrine deposits. The dune sandstones of the eolian facies show the best reservoir qualities in contrast to the inter-dune and lacustrine deposits, which show varying degree of relatively inferior quality. The 'Unayzah-A is overlain by basal Khuff-D clastics (BKDC) which is composed of medium to coarse-grained sandstones of fluvial and estuarine origin with small amount of mudstone. Even though the BKDC is identified by Area Exploration as 'Unayzah-A1, its depositional make up is different than 'Unayzah-A, and should be evaluated differently. The complex vertical and lateral interrelationships between these facies resulted in a stratified and highly heterogeneous reservoir architecture.

Reservoir Characterization

'Unayzah A eolian sandstone in this field is a complex and heterogeneous reservoir that needs to be evaluated thoroughly prior to undertaking field development. Heterogeneity which is a variation in physical and chemical properties of a reservoir rock, generally exerts a negative effect on fluid flow through a reservoir.⁵ The goal of integrated reservoir management via reservoir characterization is to understand the types of heterogeneity that affect reservoir of interest. For sandstones, heterogeneity is a function of sedimentologic characteristics and diagenetic modification of the sandstone.⁶ The fact that porous dune sandstone of the 'Unayzah A reservoir in this field is intercalated with playa, paleosol, or sandsheet, makes it very heterogeneous and compartmentalized.

Core and log data describe the rock architecture and rock properties of the 'Unayzah A reservoir as presented in Figures 2, 3. Figure 2 represents a core that is part of a larger core of the 'Unayzah A reservoir. Core sections inspected for W-1 well (Figure 4) shows fairly heterogeneous reservoir. The core exhibits high porosity, soft and short sand interval and overlays the relatively competent, thick, and low porosity sand interval below. The sandstone exhibits excellent examples of eolian ripple strata and sandflow cross strata. Phases of diagenesis were deduced from petrographic and scanning electron micrography (SEM) analysis (Figures 5 and 6). Log and core data indicated that both W-1 and W-7 wells have better reservoir quality than W-5 and W-6 wells. Special core study indicated the presence of clays in the pore structure of the 'Unayzah A reservoir. Illite clays, contacted by incompatible fluids such as undersaturated brines or fresh

water, can become mobile, thereby reducing permeability and causing formation damage. Calcium Chloride brine is incompatible with all clay types including illite. Chlorite is sensitive to low pH fluids. Acids and low-pH water can remove iron from iron-rich chlorite, allowing the clay to disperse. In addition, the iron can precipitate as insoluble iron hydroxides (Figure 6). DST test conducted on two delineation wet wells W-3 and W-4 indicated that water is unable to flow naturally across 'Unayzah-A reservoir. Presence of illite in the 'Unayzah-A sand along with high capillary bound water could be the reason why water does not flow naturally in the field.

'Unayzah-A reservoir appeared to have one pressure system based on the estimation and interpretation of reservoir pressures from DST tests on several wells throughout the field. This support the fact that the field is structurally controlled and the phenomenon of the stratigraphic control and compartmentalization are less likely to occur (Figure 7).⁷ The intersection point between the gas gradient (0.13 psi/ft) and water gradient (0.49 psi/ft; 200k ppm) indicates the gas-water contact in the field, as shown in Figure 7.

A fingerprinting study of several condensate samples from the 'Unayzah Reservoir of three 'Unayzah A wells were conducted (Figure 8).⁸ This is a useful tool to determine the continuity of the reservoir among these wells and determine if it is one simple system, or if reservoir compartments exist. Fingerprint results⁸ for each of the condensates from W-1 and W-7 are essentially identical, suggesting they are all from the same reservoir compartment. The W-6 well condensate shows significant differences from both W-1 and W-7, suggesting a separate reservoir compartment in W-6. The ratios that show the greatest differences are between C8 and C10, suggesting other factors, such as density stratification or excessive phase separation during testing (SBHP <P_d) at the field separator, and may explain the differences at W-6 well. More information is required to evaluate these possibilities.

Fracture Mechanics

Hydraulic fracture occurs when hydraulic pressure adjacent to the subsurface formation overcomes the least in-situ principal stress and initiates a fracture in a plane perpendicular to the least principal earth stress (Figure 9). A good rule of thumb is that in wells deeper than 2,000 ft (610 m), in a tectonically relaxed area, most fractures will be vertical. The direction of the induced fracture will depend entirely on the relative magnitude of the two principal horizontal stresses in the well bore, and the fracture will propagate in the direction of the maximum principal stress.

The effectiveness of fractures is dependent on proppant placement and fracture conductivity. It should be noted that formation permeability k is the most important parameter to be known prior to hydraulic fractures. High k requires a fractured well to have high conductivity for stimulation to be effective.

The location of high stress layers such as shales is also important in fracturing. If the productive formation is effectively isolated between two massive shales that may serve as barriers to the vertical extension of the fracture, then the

hydraulic fracture will be confined. The pad and the proppant can be contained within the productive zone, and the treatment will be successful.

Both lateral and vertical fracture growth depends on geomechanical properties, such as in-situ stress and Young's Modulus, net pressure, injection rate, pumped fluid characteristics such as viscosity, and the completion technique.⁹

Pre- and Post-Hydraulic Fracture Treatment

The 'Unayzah A sand in this field is highly friable and unconsolidated (Figure 4), thus has high tendency of sanding during production. Therefore, sand control completion is necessary for all wells producing from this field. The main objective of the hydraulic fracturing in this field is largely to eliminate sand production and enhance the well's deliverability. Saudi Aramco applies two methods of completions in the field: a) fracpack completion, b) indirect proppant fracture completion. The first one is combination of a hydraulic fracture treatment and gravel pack completion with the goal of creating a short, wide, high permeability fracture. This reduces the pressure drop around the wellbore to improve production, and then a mechanical screen is placed to eliminate sand production. The indirect fracture also known as the "screenless" completion, is a conventional fracture in the consolidated section that grows into the soft sand section.¹⁰ Resin coated sand is usually used as a tail-in for proppant flowback control.

If soft sand is intermixed with a competent sand interval and its thickness exceeds 20 to 30 ft in a single section, then indirect fracturing can be applied. However, in the lack of a competent section, the fracpack approach assures better control in eliminating sand.

'Unayzah W-1 Well Stimulation

The first well to be stimulated with fracpack technology was W-1 well. Figure 10 shows the reservoir properties of the 'Unayzah W1 example, with the formation lithology shown in the left column, porosity profile, PLT, and fracture conductivity in the next three columns, respectively. This well is very prolific and it was tested (prior to fracpack), with sweet gas at a rate of 40 MMSCFD with 65 bbls/MMSCFD condensate yield through 3 1/2 inch tubing at a FWHP of 2,242 psig and 64/64" choke setting. A fracpack completion across 100' of perforated 'Unayzah-A reservoir has been successfully installed in A-1 with a total of 175,000 lbs of proppant placed inside the formation. The well was flowed at a sand-free gas rate of 60 MMSCFD with no water at 3,870 psig FWHP. A post-job PLT (Figure 10) shows dominant and clear gas contributions from the upper perforated/screened interval (a streak at the top 8-10'). The fracture geometry obtained by actual pressure history match shows a full vertical coverage in the net pay section (Figure 11).¹¹ Due to the higher than expected fluid efficiency, the fracture growth was more than originally predicted and thus the final net pressure and fracture width values were lower than designed. A net

pressure of about 2,100 psi (Figure 12) was achieved toward the end of the treatment indicating excellent near-wellbore proppant packing, which is required for increased conductivity and solid flowback control. Although, the GWC seems to be positioned in the middle of the perforation interval of this well, the production was water free, indicating that gas is the only movable phase in the reservoir.

The effect of non-Darcy flow on rate performance has been investigated and quantified during the multiple rate and shut-in tests (MIT) conducted on W-1. Figure 13 presents the skin effects computed from the buildup analysis subsequent to the production period.¹¹ The negative skin achieved from the fracture treatment may be masked by the non-Darcy flow effects at high rates. The non-Darcy flow effects may be due to gas flow inside the fracture toward the wellbore, in the casing/production annulus which is packed with proppant, and in the mechanical screen section as gas enters the production string.

'Unayzah W-5 Well Stimulation

This well was drilled in the north-eastern margin of the field to delineate and develop the 'Unayzah reservoir in the field. This is a crestal well and has a shallower structural elevation than all the other wells, including the W-1 well. Since the well has a limited pay that overlays competent, thick, and low porosity sand, the well was a candidate for screenless completion. The well was tested after completing the first screenless proppant fracture treatment in field (Figure 14) and produced a sand-free gas rate of 17 MMSCFD with no water at 3,829 psig wellhead pressure. A total of 200,000 lb of proppants (ISP& RCP) was successfully placed in the formation during the operation. This well encountered sanding and proppant flowback at a high-rate of production. It appears that some of the proppants did not hold in-place after closure. The well will thus be produced at a stabilized sand-free rate, slightly lower than the tested rate, to ensure sand-free production. In order to produce at a higher rate, the well needs to be recompleted with fracpack and a mechanical screen.

Completion Design Modification

Based on the evaluation of initial completion methods conducted on wells W-1 and W-5, Saudi Aramco Stimulation Network (StimNet) team has modified the completion design for future wells. The modification includes:

1. Reduce the length of the perforated interval to only across the porous, high productive sand and avoid perforating the unproductive high clay-bound sand.
2. Reduce the pad volume to limit fracture extension prior to tip screenout and avoid the screenless method. Reduction in pad also implies less contamination of the reservoir by fracturing fluids.

The modified design proved to be effective in the other wells such W-6 and W-7. A successful fracpack operation was conducted on W-6, near W-5, and the W-6 tested sand-free gas at a rate of 18 MMSCFD with no water and 990 BPD

of condensate at a FWHP of 2,976 psig. It had no sanding problems at a higher rate, as was experienced in W-5's screenless completion.

The reservoir quality and log signature for W-7 was similar to the W-1. Lessons from W-1 were thus considered prior to fracpack design of W-7. The perforation interval has been limited to the most productive zone observed in W-1. A fracpack completion across 30' of perforated 'Unayzah-A reservoir was successfully installed in W-7. A total of 70,500 lbs of 16/30 CARBO-Prop was pumped at an average rate of about 30 BPM. A post-job pressure match was performed to determine the geometry and conductivity of the fracture. The results of the pressure match indicated a 100 ft fracture half-length with 5,000 md-ft conductivity and a fluid efficiency of about 40% (Figure 15). The well was flowed up to an estimated gas rate of 36 MMSCFD at a FWHP of 4,030 psig. No solid production was observed during the test. The well was then ramped down and shut-in with a final SIWHP of 5,380 psig. W-7 well has less potential than the nearby W-1 but tested significantly better than either of the other two wells, W-5 and W-6.

Figure 16 shows the post-frac clean-up performance for all four wells. It is clear that W-1 has the best performance of all, which could be attributed to the better reservoir quality of W-1 compared to other wells.

Conclusion

1. Saudi Aramco's use of an integrated reservoir management strategy has optimized hydraulic fracturing and post-frac well performance in this field.
2. All available pieces of documents including test, core, logs, structural analysis, condensate fingerprinting, and DST pressure buildup data, etc. were used to characterize the degree of heterogeneity and compartmentalization in the field.
3. Core and log data show various facies with different reservoir quality within the 'Unayzah A reservoir.
4. DST test results of two downdip wet wells, W-3 and W-4, across 'Unayzah-A reservoir showed that water does not flow naturally. Illite presence in the 'Unayzah-A sand along with high capillary bound water could be the cause of water not flowing naturally in the field.
5. Certain lithological sequences, such as the upper high streak in W-1, require fracpack completion over any other type of completion for sand control and high, sustained rates. The completion requirement would be different for competent, low porosity, sand intercalated with high amounts of clay-type intervals.
6. Screenless completion was not fully effective in this field, while fracpack completion was more effective way for development.

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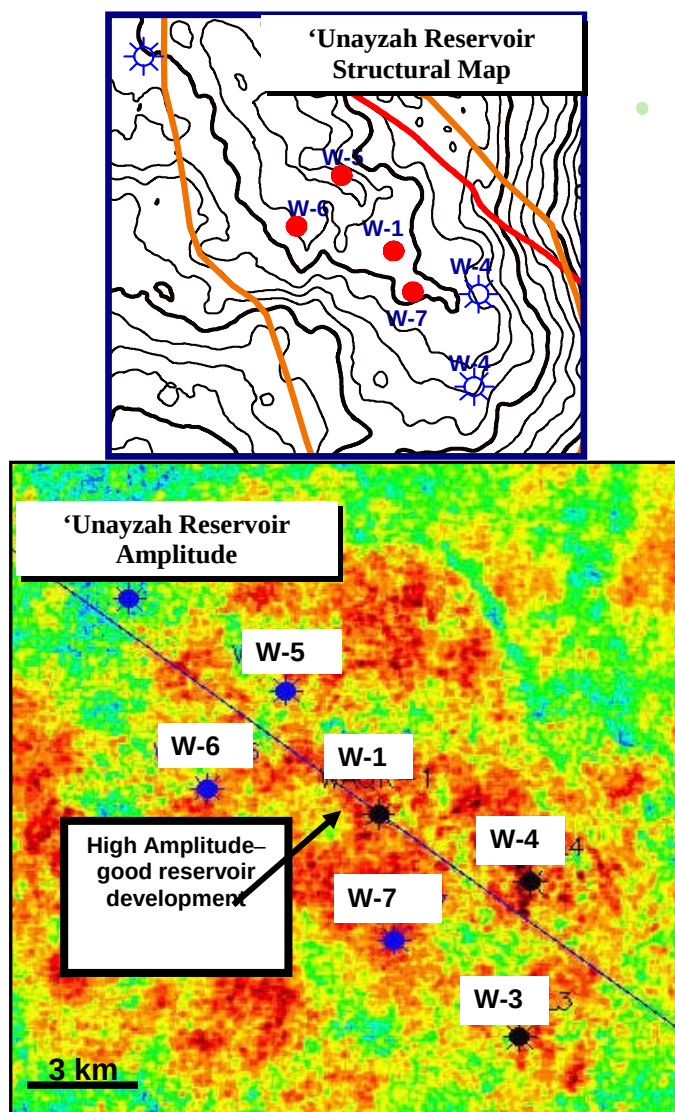


Figure 1. Structural and 3-D seismic maps of the 'Unayzah A reservoir

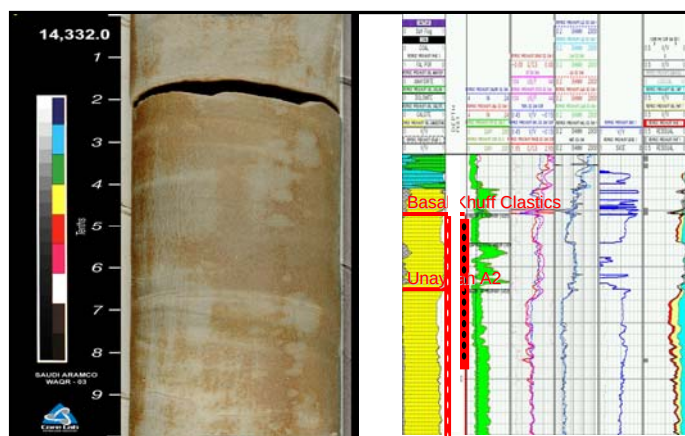


Figure 2. Whole Core Photograph of a typical 'Unayzah A gas well.

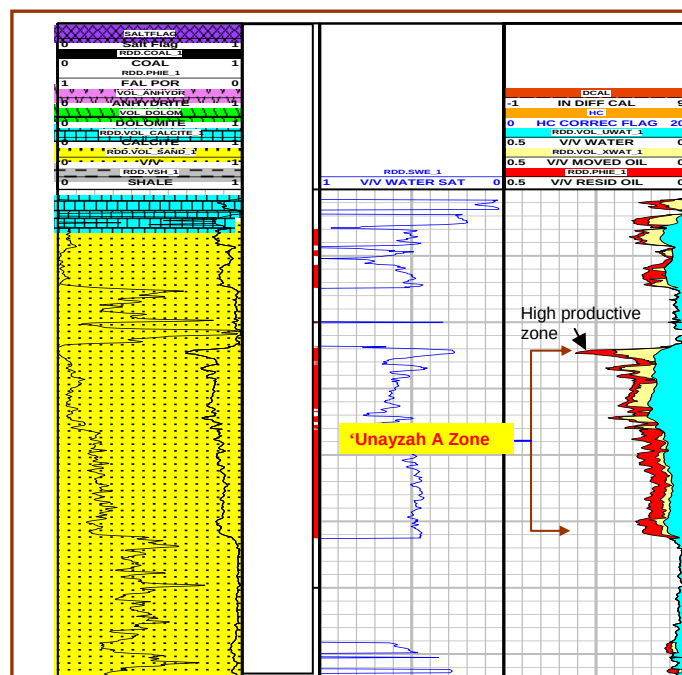


Figure 3. Typical porosity and lithology log of 'Unayzah A.

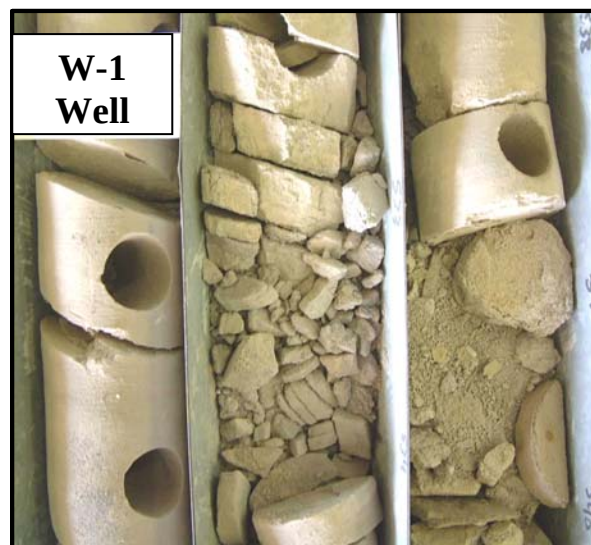


Figure 4. Unconsolidated 'Unayzah A core, W-1.

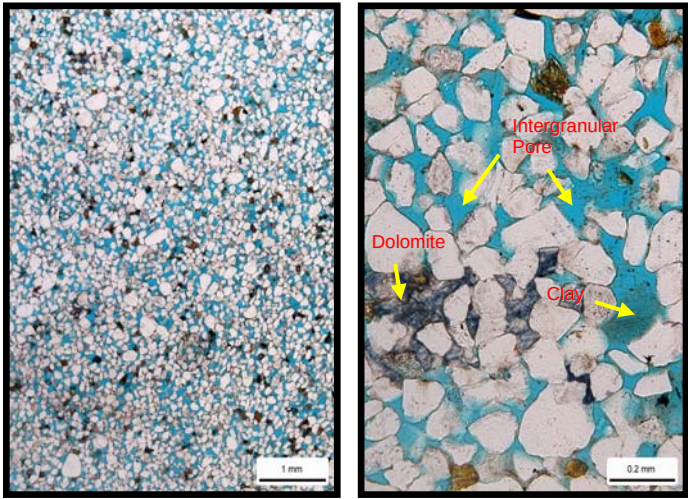


Figure 5. Thin Section Photomicrograph: 'Unayzah A dolomite cement, pore-filling clay, yet good apparent porosity.

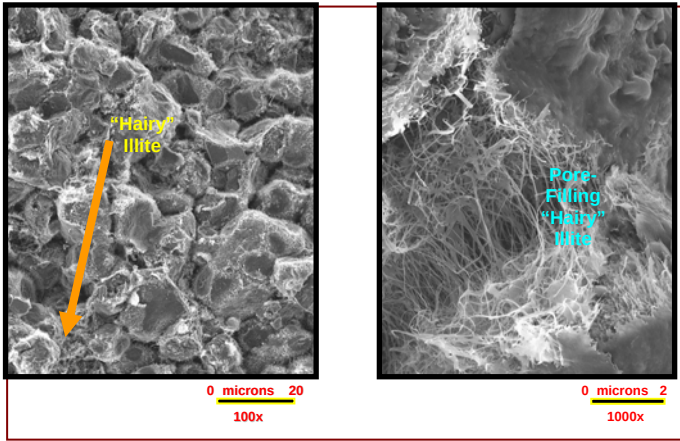


Figure 6. SEM Photomicrograph: Unayzah A2 remobilization of pore-filling "hairy" Illite by drlg / comp Fluids.

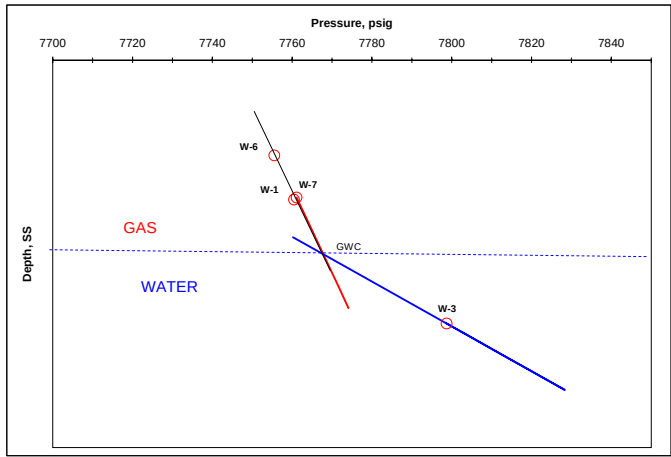


Figure 7. Pressure profile of the field estimated from buildup tests conducted at the end of DST tests.

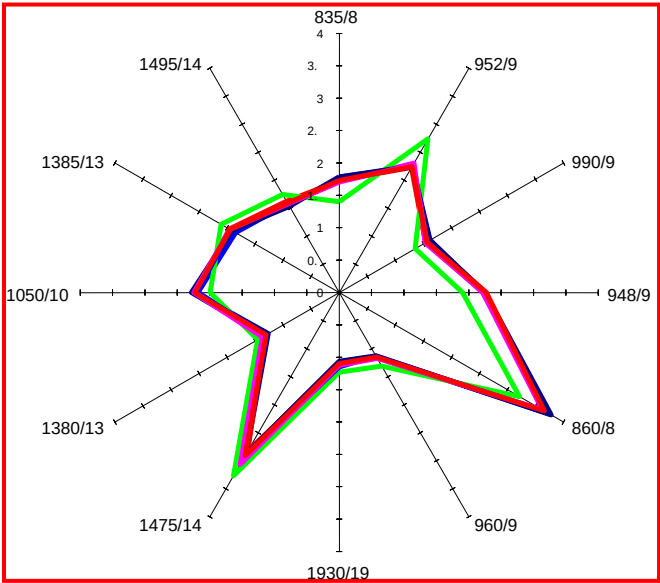


Figure 8. Fingerprint results for condensates from W-1 (red) and W-6 (green) and W-7 (blue).⁸

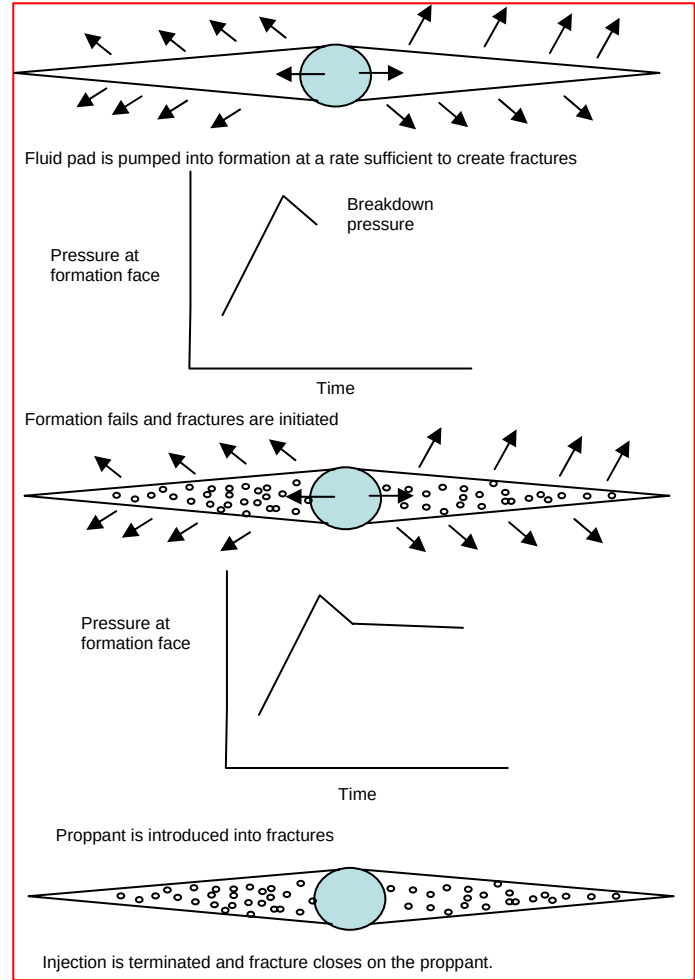


Figure 9. Schematic of fracture mechanics.

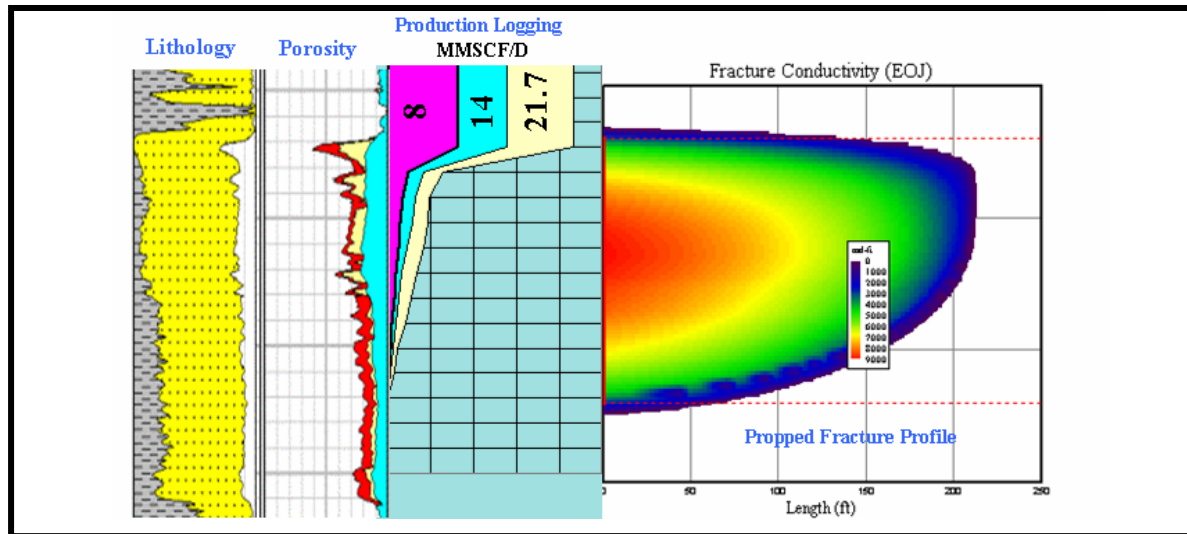


Figure 10. Reservoir properties, fracture dimensions, and PLT for W-1 'Unayzah.¹¹

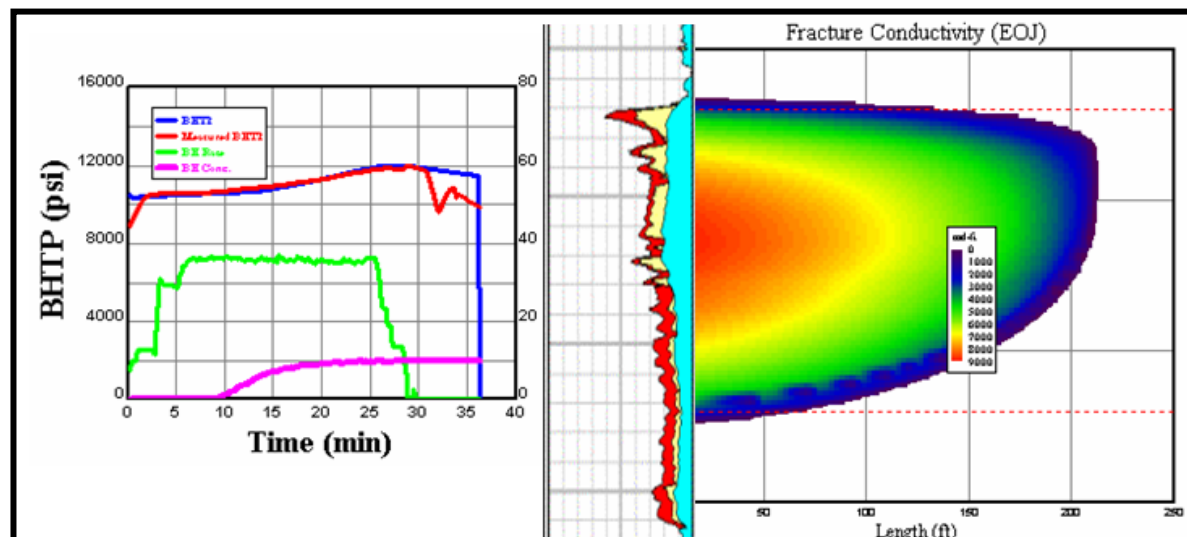


Figure 11. Computed geometry using actual pump schedule and history match of pressure data.¹¹

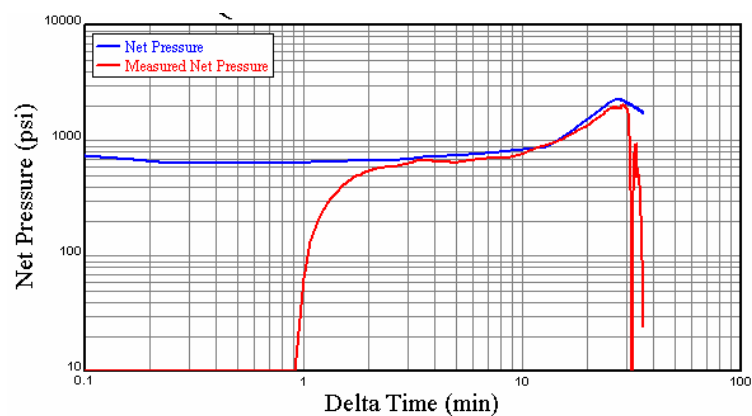


Figure 12. Net pressure match for the main fracture treatment, W-1.¹¹

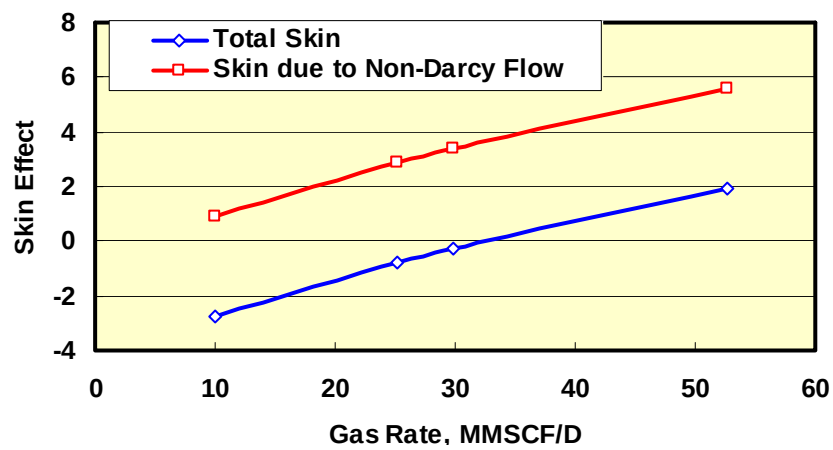


Figure 13. Total and non-Darcy skin estimation during multi-rate testing, W-1.¹¹

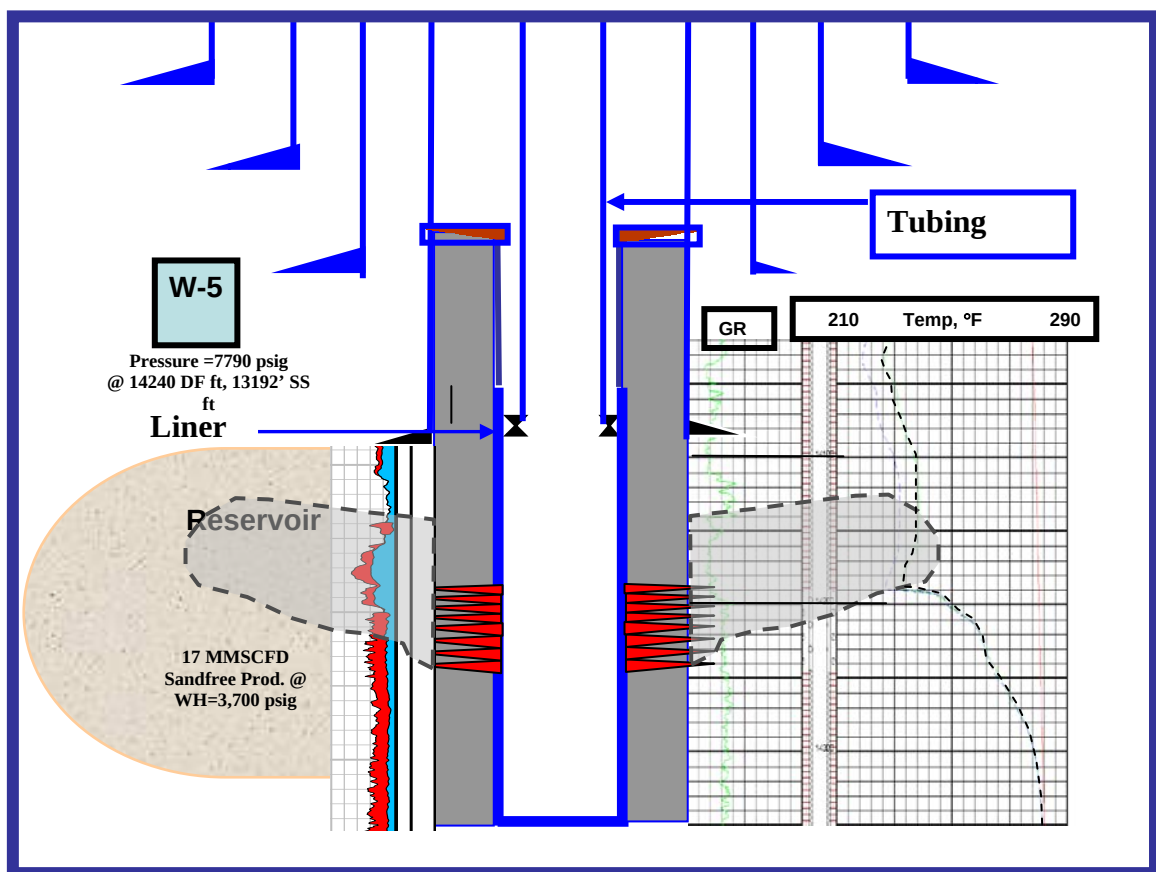


Figure 14. Schematic of reservoir properties and screenless fracture design for W-5 'Unayzah A.

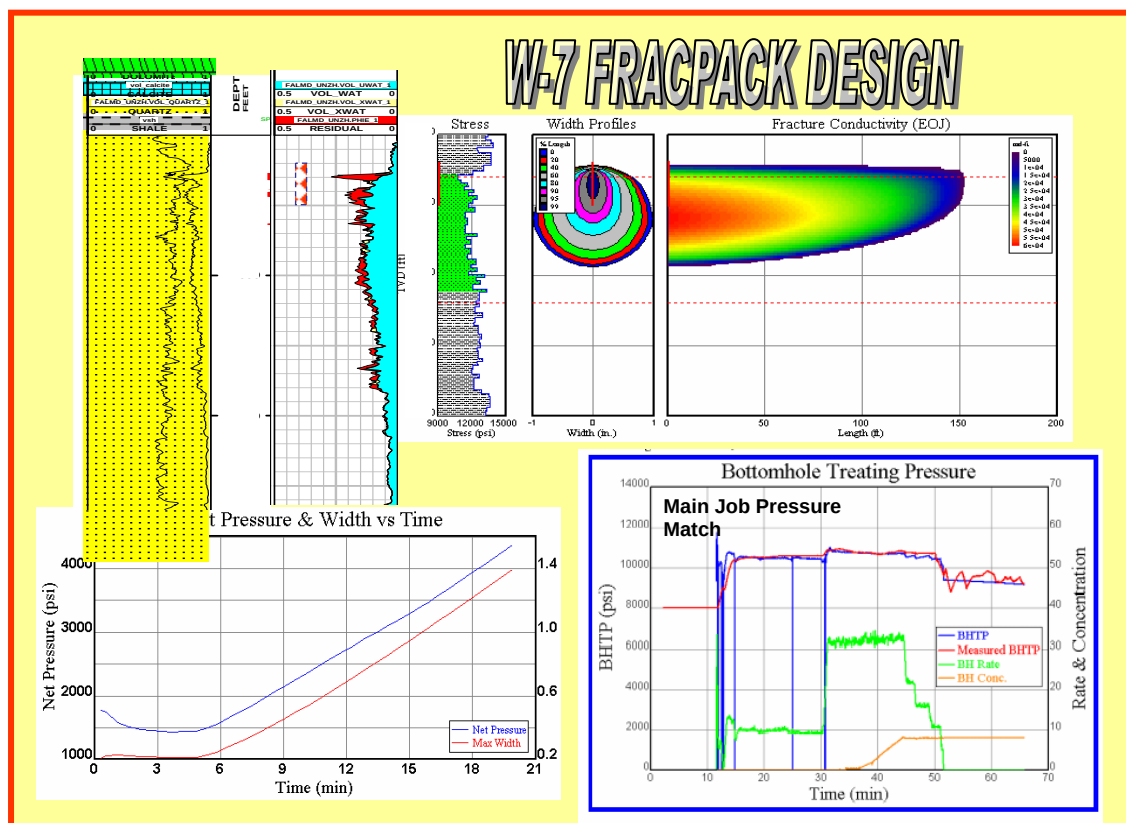


Figure 15. Net pressure match for the main fracture treatment, well W-7.

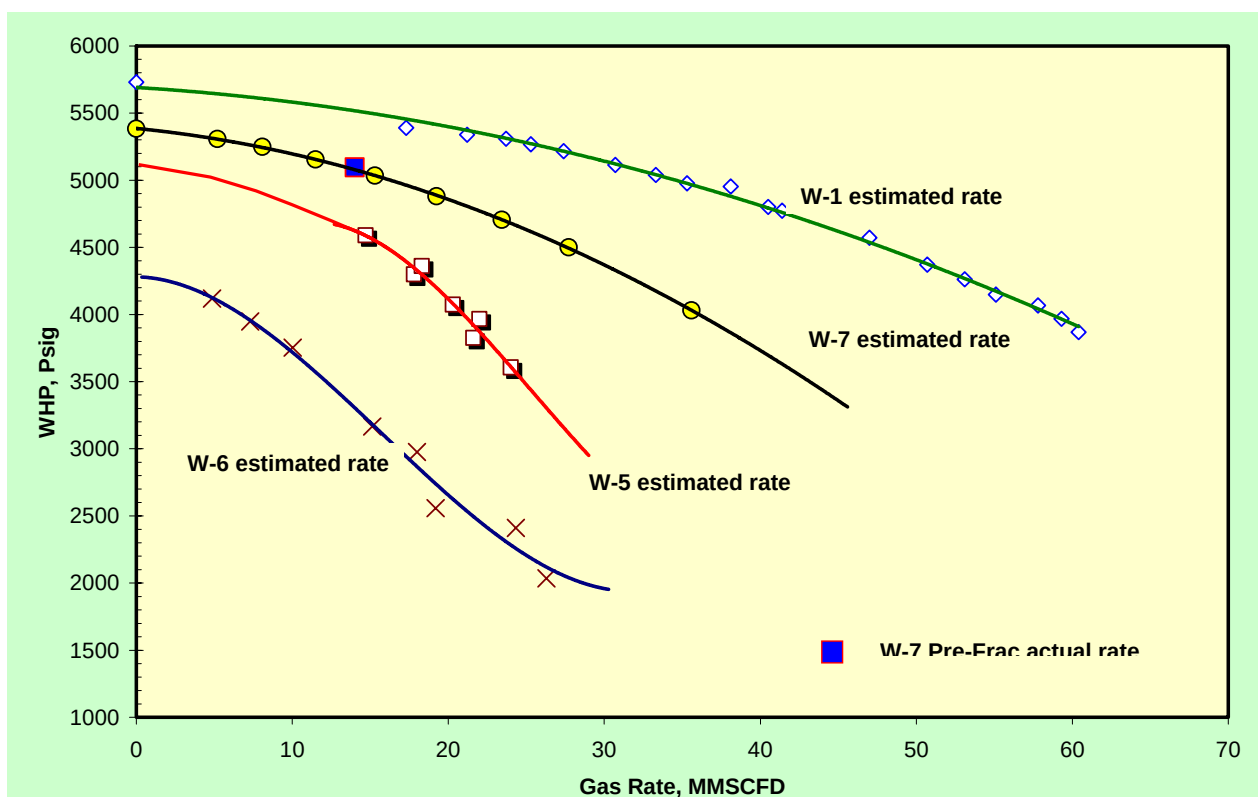


Figure 16. Post-Frac Initial Performance for all wells.